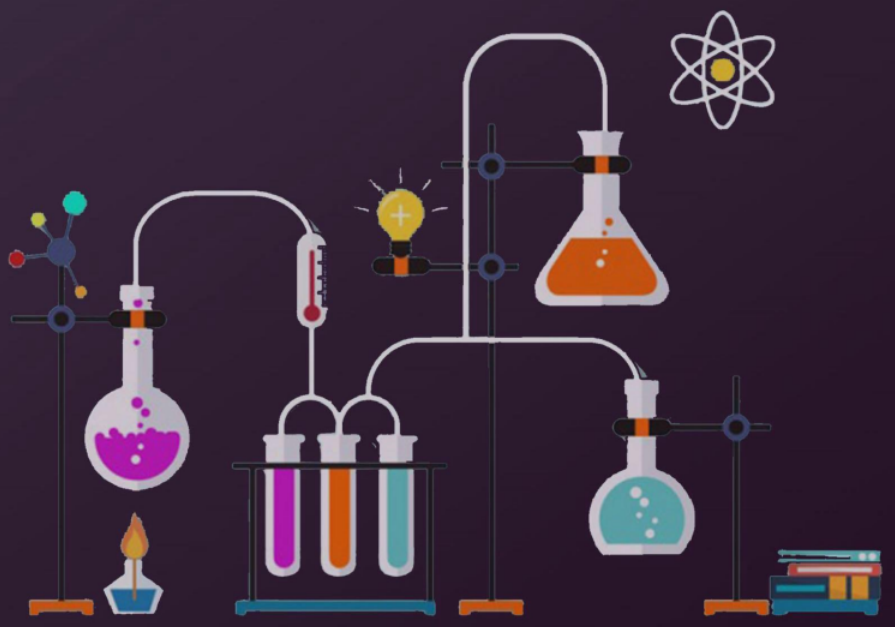
CHEMISTRY

p-Block Group (13-14)



SHEET

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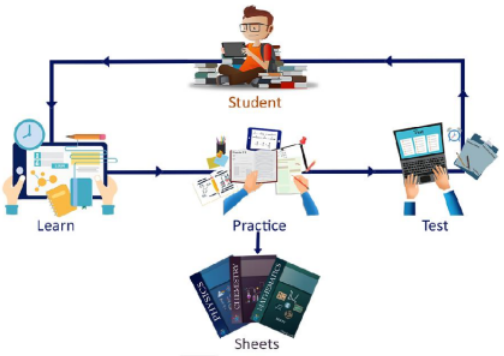
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction to P-Block Elements	1,2			
2	Boron Family	3			
3	Chemical Properties of Boron Family	4,5			
4	Preparation of Boron and it's compounds	6,7,9,20	1,13	2,3	1,4,6,12
5	Some Important Compounds and uses of Boron and Aluminium	8, 34	18		2
6	Carbon Family and its chemical properties	10,11	4		
7	Carbon and it's allotropes	12	7		
8	Some Important compounds of Carbon and Silicon	13,14,17	5,10		
9	The electronic element: Silicon!		3,8,9	4,6	
10	Silicates	15,18,29	15		
11	ACC	16,19-36	2,6,11,12,14, 16,17,19	1,5,7,8	3,5,7-11

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction to P-Block Elements

Q1. Al and Ga have nearly same covalent radii because of :-
(A) Poor shielding power of d-electrons of Ga atoms
(B) Greater shielding power of d-electrons of Ga atoms
(C) Greater shielding power of s-electrons of Ga atoms
(D) Poor shielding power of s-electrons of Ga atoms

Q2. Which of following is Acetylide ion :
(A) C^{2-} (B) C_2^- (C) C_2^{2-} (D) C_3^{4-}

Chemical Properties of Boron Family

Q3. The main factor responsible for weak acidic nature of BF_3 is :-
(A) Large electronegativity of F
(B) 3c - $2e^-$ bond in BF_3
(C) $p\pi - d\pi$ back boding in B – F
(D) $p\pi - p\pi$ back bond in BF_3

Q4. Thallium shows different oxidation states due to :-
(A) Inert pair effect
(B) Amphoteric nature
(C) Transition metal
(D) High reactivity

Q5. The state of hybridisation of boron in BCl_3 is :-
(A) sp (B) sp^2 (C) sp^3 (D) sp^3d^2

Preparation of Boron and it's important compounds

Q6. Which of the following have 3c–$2e^-$ / banana bond ?
(A) Al_2Cl_6 (B) $BeCl_2$
(C) B_2H_6 (D) All of these

Q7. Glass is soluble in :-
(A) HF (B) H_2SO_4
(C) $HClO_4$ (D) aquaregia

Q8. Inorganic Benzene and Inorganic graphite are respectively :
(A) $B_3N_3H_6$, BN (B) BN, $B_3N_3H_6$
(C) B_2O_3, BN (D) BN, B_2O_3

Q9. When B_2H_6 Reacts with Ammonia, it gives:-
(A) $[H_2B(NH_3)_2]^+ + [BH_4]^-$
(B) $[H_4B(NH_3)_2]^- + [BH_2]^+$
(C) $2BH_3(NH_3)_2$
(D) No reaction

Carbon Family and its Chemical Properties

Q10. Which of the following doesn't exist :-
(A) BF_6^{3-} (B) AlF_6^{3-}
(C) SiF_6^{2-} (D) $GeCl_6^{2-}$

Q11. The stability of dihalides of group 14 elements follow which of the order ?
(A) $SiCl_2 < GeCl_2 < PbCl_2 < SnCl_2$
(B) $GeCl_2 < PbCl_2 < SnCl_2 < SiCl_2$
(C) $SiCl_2 < GeCl_2 < SnCl_2 < PbCl_2$
(D) $GeCl_2 < SiCl_2 < PbCl_2 < SnCl_2$

Q12. Consider the following chemical reaction :

$$\underset{\text{(Formic acid)}}{HCOOH(aq.)} \xrightarrow{X,\Delta} CO(g) + H_2O$$

In the above reaction (X) is –
(A) Conc. H_2SO_4 (B) Conc. HCl
(C) Conc. HNO_3 (D) Dil. HNO_3

The electronic element: Silicon!

Q13. Silicone have general formula :

(A) SiO_4^{4-} (B) $Si_4O_7^{6-}$

(C) $(R_2SiO)_n$ (D) $(SiO_3)_n^{2-}$

Q14. Which of the following oxides has a three dimensional structure ?

(A) CO (B) CO_2

(C) SiO_2 (D) SO_2

Silicates

Q15. Name the type of the structure of silicate in which one oxygen atom of $[SiO_4]^{4-}$ is shared?

(A) Linear chain silicate

(B) Sheet silicate

(C) Pyrosilicate

(D) Three dimensional silicate

Q16. Zeolite can be used as :-

(A) Catalyst

(B) In water softning

(C) In making of glass

(D) Both (A) and (B)

Q17. Carborundum is :-

(A) SiO_2 (B) SiF_4

(D) Si_3N_4 (D) SiC

Q18. The Basic building unit of silicates is :-

(A) SiO_6^{2-} (B) SiO_4^{4-}

(C) SiO_2^{6-} (D) SiO_6^{2+}

ACC

Q19. SnO_2 is :-

(A) Acidic (B) Alkaline

(C) Amphoteric (D) Neutral

Q20. Which is not true about borax?

(A) It is a useful primary standard for titrating against acids.

(B) One mole of borax reacts with four moles of HCl.

(C) Aqueous solution of borax can be used as a buffer.

(D) It is made up of two triangular BO_3 units and two tetrahedral BO_4 units.

Q21. An inorganic compound (A) made of two most occurring elements into the earth crust, having a polymeric tetra-headral network structure. With carbon, compound (A) produces a poisonous gas (B) which is the most stable diatomic molecule. Compounds (A) and (B) will be :

(A) SiO_2, CO_2 (B) SiO_2, CO

(C) SiC,CO (D) SiO_2, N_2

Q22. Which of following molecule can show Lewis acidity.

(I) CO_2 (II) Br_2

(III) $SnCl_2$ (IV) HF

(V) NMe_3

(A) III, IV (B) I, II, III

(C) I, III, IV (D) II, III, V

Q23. When conc. H_2SO_4 was treated with $K_4[Fe(CN)_6]$, CO gas was evolved. By mistake, somebody used dilute H_2SO_4 instead of conc. H_2SO_4 then the gas evolved was :

(A) CO (B) HCN

(C) N_2 (D) CO_2

Q24. Dangling valency is not exhibited by which of the following allotrope of carbon ?

(A) Graphite (B) Diamond

(C) C_{60} (D) All of these

Q25. $RCl \xrightarrow[Si]{cu-powder} R_2SiCl_2 \xrightarrow{H_2O}$

$R_2Si(OH)_2 \xrightarrow{condensation} A$

Compound (A) is :-

(A) a linear silicone (B) a chlorosilane

(C) a linear silane (D) a network silane

Q26. The coordination number of $AlCl_3$ in their solid, liquid and gaseous state is respectively:

(A) 6,6,4 (B) 6, 4, 4

(C) 6,6,6 (D) 4,4,4

Q27. In Which of the following hybridisation state of the central atom would be changed if transformed from monomeric to dimeric or polymeric species.

$BeCl_2(g) \rightarrow BeCl_2(s)$

$AlCl_3(g) \rightarrow Al_2Cl_6(l)$

$BeH_2(g) \rightarrow BeH_2(s)$

$ICl_3(g) \rightarrow I_2Cl_6$

(A) sp^2 to sp^3 (B) sp to sp^3

(C) sp^3d to sp^3d^2 (D) All the above

Q28. Which one of the following statements is not true regarding diborane?

(A) It has two bridging hydrogens and four perpendicular to the rest.

(B) When methylated, the product is $Me_4B_2H_2$.

(C) The bridging hydrogens are in a plane perpendicular to the rest.

(D) All the B–H bond distances are equal.

Q29. In which of the following silicates, only two corners per tetrahedron are shared :

(i) Pyrosilicate

(ii) Cyclic silicate

(iii) Double chain silicate

(iv) Single chain silicate

(v) 3 D silicate

(vi) Sheet silicate

(A) (i), (ii) and (v) (B) (iv) and (vi) only

(C) (i) and (vi) only (D) (ii) and (iv) only

Q30. (a) $Al \xrightarrow{N_2} A$, (b)$Al \xrightarrow{C} B$, Product A and B on hydrolysis yields respectively.

(A) Ammonia and acetylene

(B) Ammonia and methane

(C) Nitric oxide and acetylene

(D) None

Q31. Which gas is responsible for green house effect :

(A) CO_2 (B) SO_2

(C) CO (D) SO_3

Q32. Artificial gem used for cutting glass is :

(A) Graphite (B) Diamond

(C) SiC (D) $CaCN_2$

Q33. Which of the following properties describes the diagonal relationship between boron and silicon?

(A) BCl_3 is not hydrolysed while $SiCl_4$ can be hydrolysed

(B) Both form oxides; B_2O_3 is amphoteric; SiO_2 is acidic

(C) Both metals dissolve in cold and dilute nitric acid

(D) Borides and silicides are hydrolysed by water

Q34. Aluminium vessels should not be washed with materials containing washing soda because :-

(A) Washing soda is expensive.

(B) Washing soda is easily decomposed

(C) Washing soda reacts with aluminium to form soluble aluminate

(D) Washing soda reacts with aluminium to form insoluble aluminium oxide.

Q35. Borax is converted into amorphous Boron by following steps :

$$\text{Borax} \xrightarrow{X} H_3BO_3 \xrightarrow{\Delta} B_2O_3 \xrightarrow[\Delta]{Y} B$$

X and Y are respectively.

(A) HCl, Mg (B) HCl, C

(C) Δ, Al (D) HCl, Al

Q36. Borax bead test not applicable for :–

(A) Cu (B) Mn

(C) Zn (D) Al

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Select correct statement about B_2H_6 :-

(A) Bridging group are electron-rich with 12 valence electrons.

(B) It has 2c - 2e B–H bonds.

(C) It has 3c - 2e B–H bonds.

(D) It has sp^3 hybridised boron atoms.

Q2. Which of the following statements is incorrect :-

(A) $SnCl_4$ is ionic and is strong reducing agent.

(B) $SnCl_4$ is covalent.

(C) $PbCl_4$ is reducing agent.

(D) Lower valencies are more ionic because of the greater radius of M^{2+} than of M^{4+}.

Q3. Which of the following are true about silicones-

(A) They are formed by the hydrolysis of R_2SiCl_2.

(B) They are polymer made by R_2SiO_2 units.

(C) They are made up of R_2SiO_2 unit.

(D) They are macromolecules.

Q4. Carbon differs from the rest of the family members because of :-

(A) Number of unpaired electrons in the valence shell.

(B) Small size.

(C) Non availability of vacant orbitals in valence shell.

(D) Non-availability of d-orbital in valence shell.

Q5. Which of the metal oxides are reduced by CO :-

(A) ZnO (B) Fe_2O_3

(C) CaO (D) Na_2O

Q6. $Na_2CO_3 \xrightarrow[\text{excess}]{SO_2}$ solution (A)

$\xrightarrow{NaIO_3 \text{ solution}}$ (B) (final product).

Which of the following statements is/ are correct regarding 'B'.

(A) It has no electrical conductivity in solid state but conducts electricity in liquid state.

(B) It is highly soluble in water.

(C) It produces blue solution with starch.

(D) It produces the same oxidation state on reaction with excess Cl_2 water as that of 2nd reagent used above.

Q7. Which of following statement is/are correct regarding graphite?

(A) Graphite is thermodynamically more stable than diamond.

(B) d_{C-C} among the layers and within the layers are not same.

(C) Graphite is more reactive than diamond.

(D) Graphite conducts electricity due to its bond polarity.

PARAGRAPH TYPE

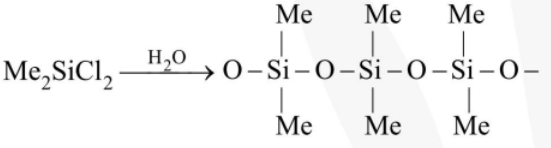
*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for question nos. 8 to 9

Read the following write-ups and answer the questions at the end of it.

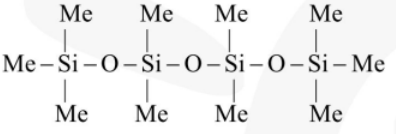
Silicons are synthetic polymers containing repeated R_2SiO units. Since, the empirical formula is that of a ketone (R_2CO), the name silicone has been given to these materials. Silicones can be made into oils, rubbery elastomers and resins. They find a variety of applications because of their chemical inertness, water repelling nature, heat-resistance and good electrical insulating property.

Commercial silicon polymers are usually methyl derivatives and to a lesser extent phenyl derivatives and are synthesised by the hydrolysis of R_2SiCl_2[R=methyl (Me) or phenyl (ϕ)] .

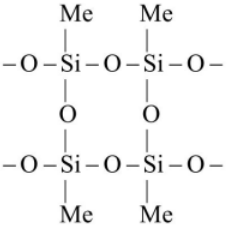


Q8. If we mix Me_3SiCl with Me_2SiCl_2, we get silicones of the type:

(A)



(B)

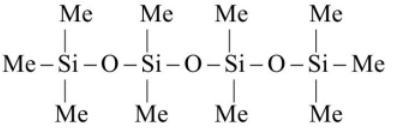


(C) both of the above

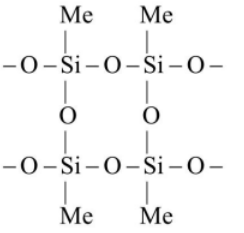
(D) none of the above

Q9. If we start with $MeSiCl_3$ as the starting material, silicones formed is:

(A)



(B)



(C) Both of the above

(D) None of the above

Paragraph for question nos. 10 to 12

The term carbide is generally applied to compounds in which carbon is bonded to the elements of lower or approximately same electronegativity. This definition excludes the compounds in which oxygen, sulphur, phosphorus nitrogen and halogens are united with carbon. Reactive metals (i.e. of gp 1 and 2) form ionic carbides. They hydrolyse to liberate hydrocarbons. Most of them resembles with NaCl in crystal structure. Transition metals forms interstitial carbides.

Q10. Consider the following Carbides -

CaC_2	Be_2C	MgC_2	SrC_2
I	II	III	IV

Select the carbide which gives different product on hydrolysis, than other carbides:-

(A) I (B) II

(C) III (D) IV

Q11. What is the co-ordination number of Ca^{2+} in solid CaC_2 ?

(A) 4 (B) 6

(C) 8 (D) 12

Q12. Select the methanides from compounds give below :-

Al_4C_3	Be_2C	MgC_2	BeC_2
I	II	III	IV

(A) I Only (B) I and IV

(C) I and II (D) I, II, III and IV

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q13. Match the column :-

Column–I (Reactions)		**Column–II (Product)**
(A) Borax	$\xrightarrow{\Delta}$	(p) BN
(B) $B_2H_6 + H_2O$	$\longrightarrow$	(q) B_2H_6
(C) $B_2H_6 + NH_3$ (Excess)	$\xrightarrow{\Delta}$	(r) H_3BO_3
(D) $BCl_3 + LiAlH_4$	$\longrightarrow$	(s) $NaBO_2 + B_2O_3$

Q14. Match the following

Column–I	**Column–II**
(A) $BBr_3 + H_2 \rightarrow B$	(p) Borax bead test
(B) $Na_2B_4O_7.10\,H_2O + CuSO_4 \rightarrow Cu(BO_2)_2$	(q) Reduction
(C) $AlCl_3 + H_2O \rightarrow HCl$	(r) White fumes
(D) $Cr_2O_3 + Al \rightarrow Cr + Al_2O_3$	(s) Hydrolysis

Q15.

Column–A	**Column–B**
(A) $(SiO_3)_n^{2n-}$	(i) Pyrosilicate
(B) $(Si_4O_{11})_n^{6n-}$	(ii) Two dimensional sheet silicates
(C) $(Si_2O_5)_n^{2n-}$	(iii) Cyclic silicates
(D) $Si_2O_7^{6-}$	(iv) Chain silicates

Assertion and Reason

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q16. **Statement-I :** $NaBO_3/OH^-$ can be used for oxidation of Cr^{3+} to Cr^{6+}.

Statement-II : In alkaline medium $NaBO_3$ produces H_2O_2.

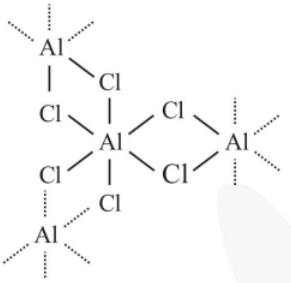
Q17. **Statement-I :** Borax bead test is applicable only for coloured salt.

Statement-II : In borax bead test, coloured salts are decomposed to give coloured metal meta borates.

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q18. The coordination number of Al in the crystalline state of $AlCl_3$ is ?



TRIPLE MATCH

*This section contains TWO tables (each having 3 columns and 4 rows). Based on each table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

	Compound	Hybridisation	Shape
(A)	SiO_4^{4-}	(i) sp^3d^2	(1) Planar
(B)	Dimer of BeH_2	(ii) sp	(2) Linear
(C)	I_2Cl_6	(iii) sp^2	(3) Square Planar
(D)	Hydrolysed gaseous product of MgC_2	(iv) sp^3	(4) Tetrahedral

Exercise - 2 *Concept Builder*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Group 13 elements show +1 and +3 oxidation states. Relative stability of +3 oxidation state may be given as :

(A) $Tl^{3+} > In^{3+} > Ga^{3+} > Al^{3+} > B^{3+}$

(B) $B^{3+} > Al^{3+} > Ga^{3+} > In^{3+} > Tl^{3+}$

(C) $Al^{3+} > Ga^{3+} > Tl^{3+} > In^{3+} > B^{3+}$

(D) $Al^{3+} > B^{3+} > Ga^{3+} > Tl^{3+} > In^{3+}$

Q2. What is not true about borax ?

(A) Molecular formula is $Na_2B_4O_7.10H_2O$

(B) Crystalline borax contains tetra nuclear unit

(C) It hydrolyses to give an acidic solution

(D) White crystalline solid

Q3. The number of three centre two electron bonds in a molecule of diborane is :

(A) 0 (B) 2

(C) 4 (D) 6

Q4. Among the following substituted silanes the one which will give rise to cross-linked silicone polymer on hydrolysis is :-

(A) R_3SiCl (B) R_4Si

(C) $RSiCl_3$ (D) R_2SiCl_2

Q5. Which of the following halide does not exist:-

(A) PbF_4 (B) $PbCl_4$

(C) CCl_4 (D) PbI_4

Q6. The product obtained when silica reacts with hydrogen fluoride at a temperature of $< 35^\circ$ C :-

(A) SiF_4 (B) H_2SiF_6

(C) H_2SiF_4 (D) H_2SiF_3

Q7. $Na_2B_4O_7 \xrightarrow{\Delta} 2NaBO_2 + B_2O_3$

Salt of Co $\xrightarrow{\Delta}$ CoO + gas

$CoO + B_2O_3 \xrightarrow{\Delta} (X)$

Find out formula and colour of X compounds.

(A) $Co(BO_2)_2$, Pink (B) $Co(BO_2)_2$, Blue

(C) $CoBO_2$, Violet (D) $Co(BO_2)_2$, Red

Q8. $RCl \xrightarrow[Si]{cu-powder} R_2SiCl_2 \xrightarrow{H_2O}$

$R_2Si(OH)_2 \xrightarrow{condensation} A$

Compound (A) is :

(A) A linear silicone

(B) A chlorosilane

(C) A linear silane

(D) A network silane

Exercise - 2A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Boric acid :-

(A) exists in layered structure form due to inter-molecular hydrogen bonding

(B) is used in manufacturing of optical glasses

(C) is a tri-basic acid

(D) with borax, it can be used in preparation of buffer solution

Q2. In Al_2Cl_6 which of the following statement is/are correct :-

(A) Four Al–Cl bonds are of same length and four of different length

(B) Six Al–Cl bonds are of same length and two of different length.

(C) The angle Cl–Al–Cl is 110° and 93°.

(D) The angle Al–Cl–Al is 87°.

Q3. Orthoboric acid (H_3BO_3) and metaboric acid (HBO_2) differ in respect of :-

(A) basicity (B) structure

(C) melting point (D) oxidation no.

Q4. Possible oxidation states of boron family elements are -

(A) + 1 (B) + 2

(C) + 3 (D) 4

Q5. $C(OH)_4$ is unstable because a carbon atom cannot hold more than one –OH groups but $Si(OH)_4$ is a stable compound because :-

(A) C–O bond energy is low

(B) C–O bond energy is high

(C) Si-O bond energy is low

(D) Si-O bond energy is high

Q6. Aqueous solution of boric acid is treated with Salicylic acid. Which of the following statements is/are **incorrect** for the product formed in the above reaction :

(A) No product will be formed because both are acid.

(B) Product is 4-coordinated complex and optically resolvable.

(C) Product is 4-coordinated complex and optically non resolvable

(D) There are two ring only which are five membered.

Q7. The non existence of PbI_4 is due to :

(A) highly oxidising nature of Pb^{4+}

(B) highly reducing nature of Pb^{4+}

(C) Sufficiently large covalent character

(D) Highly reducing power of I^-

Q8. Which of the following is/are not correct -

(A) $Ge(OH)_2$ is amphoteric

(B) $SnCl_4$ is more stable than $SnCl_2$

(C) Trisilyl amine is pyramidal

(D) $GeCl_4$ in HCl forms $H_2[GeCl_6]$

Q9. Which of the following carbides on treatment with water give methane -

(A) CaC_2 (B) Be_2C

(C) Al_4C_3 (D) Mg_2C_3

Q10. Which of the following oxides is/are neutral:-

(A) CO (B) SnO_2 (C) ZnO (D) NO

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q11. Match the following :

Column-I	**Column-II**
(A) $Bi^{3+} \rightarrow (BiO)^+$	(P) Heat
(B) $[AlO_2]^- \rightarrow Al(OH)_3$	(Q) Hydrolysis
(C) $SiO_4^{4-} \rightarrow Si_2O_7^{6-}$	(R) Acidification
(D) $(B_4O_7^{2-}) \rightarrow [B(OH)_3]$	(S) Dilution by water

Q12.

Column-A	**Column-B**
(A) Inorganic benzene	(P) $K_2SO_4.Al_2SO_4.24H_2O$
(B) Borax	(Q) B_2H_6
(C) Diborane	(R) $B_3N_3H_6$
(D) Alum	(S) $Na_2B_4O_7.10H_2O$

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
A	C	D	A	B	C	A	A	A	A
11	12	13	14	15	16	17	18	19	20
C	A	C	C	C	D	D	B	C	B
21	22	23	24	25	26	27	28	29	30
B	B	B	B	C	A	D	D	B	D
31	32	33	34	35	36				
A	C	C	D	D	D				

Ex-1A

1. B,C,D **2.** A,C **3.** B,D **4.** B,D **5.** A,B

6. A,C,D **7.** A,B,C **8.** A **9.** B **10.** B

11. B **12.** C

13. (A) s (B) r (C) p (D) q **14.** (A) q (B) p (C) r, s (D) q

15. (A) → (iii),(iv) ; (B) → (iv) ; (C) → (ii) ; (D) → (i)

16. A **17.** A **18.** 6

19. (A) – iv – 4, (B) – iii – 1, (C) – i – 3, (D) – ii – 2

Ex-2							
1	2	3	4	5	6	7	8
B	C	B	C	D	B	B	A

Ex-2A

1. A,B,D **2.** A,C,D **3.** [A,B,C] **4.** [A,B,C] **5.** A, D

6. A, C, D **7.** [A,D] **8.** C **9.** [B,C] **10.** [A,D]

11. (A) Q,S (B) R (C) P (D) Q,R **12.** A → R; B→ S; C → Q; (D) → P